

LABORATORY OF AUTOMATION SYSTEMS

Davide Samorì

Dipartimento di Ingegneria dell'Energia Elettrica e dell'Informazione (DEI)

Università di Bologna

Email: davide.samori@unibo.it

BASE MODEL FOR TRACING

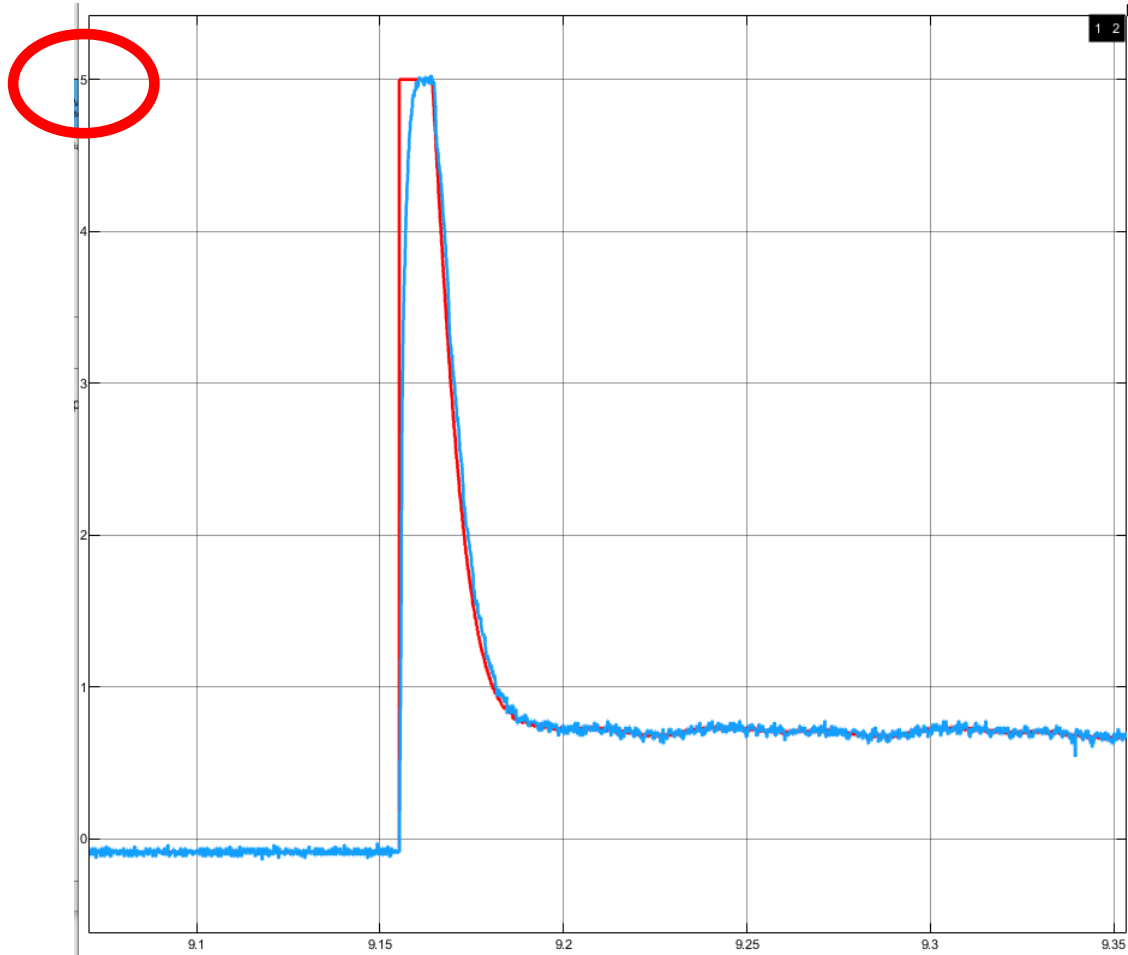
Today Goal:

- Graph signals

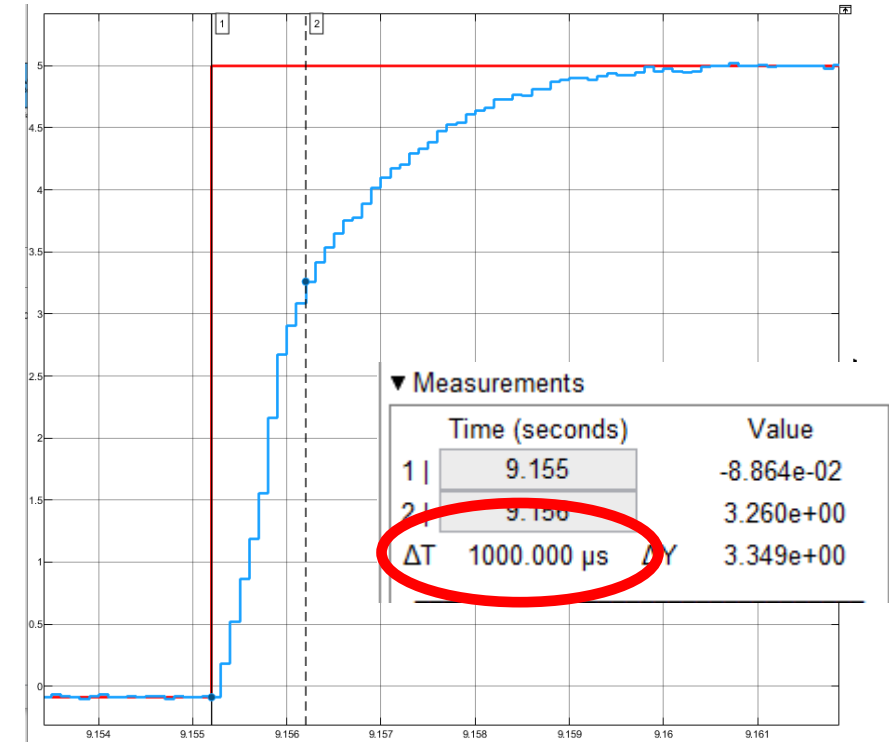
You can open the model made in the last lesson

Why do we need graphs?

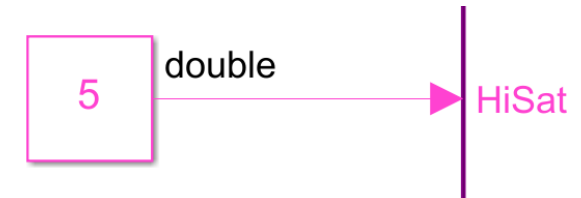
SOME TRACING EXAMPLE:



I_{REF}
 I_{MIS}

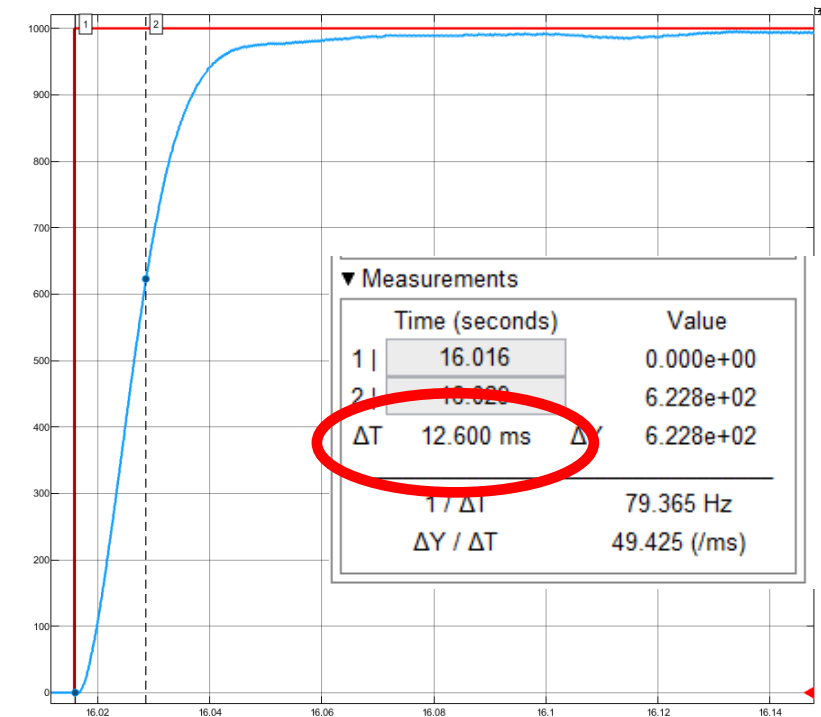
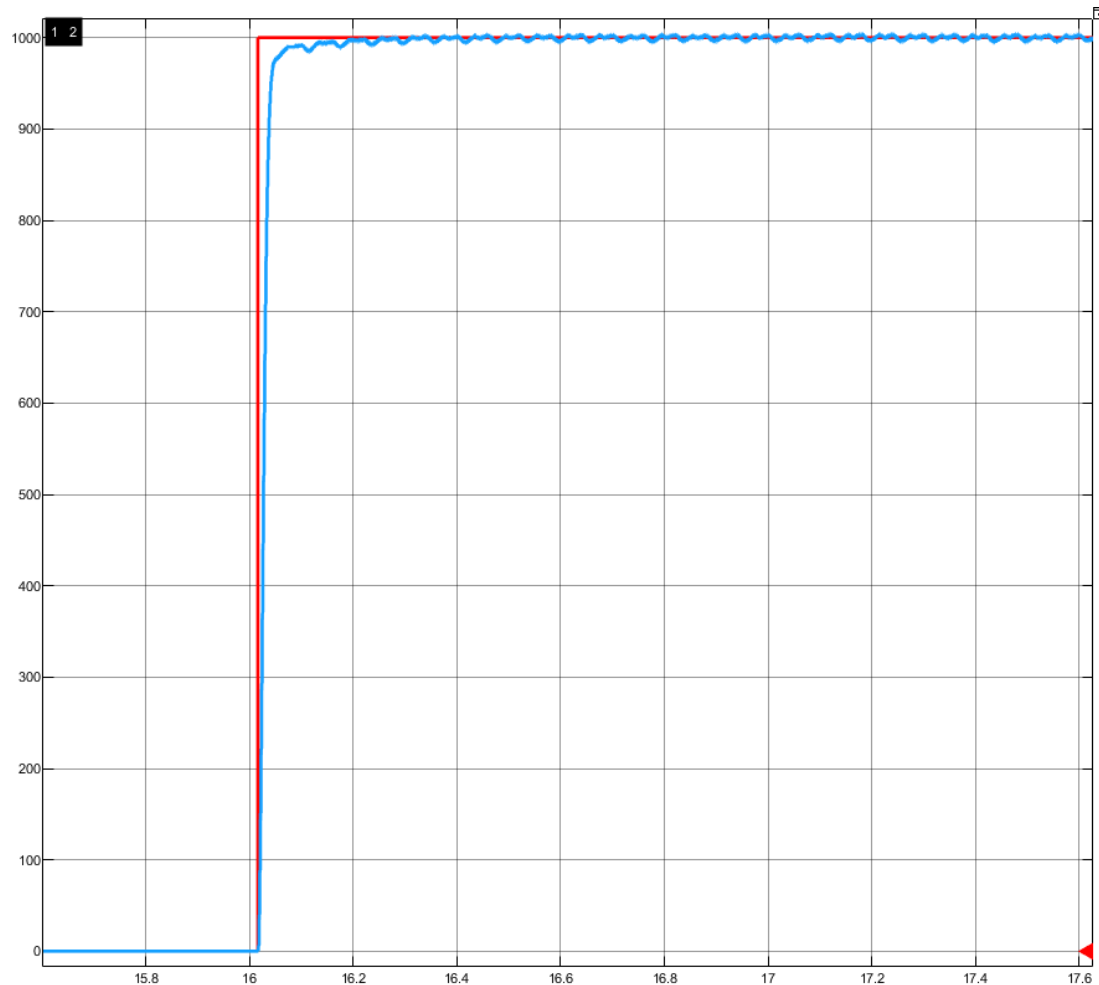


%Desired Taol
Taol=1e-3;



SOME TRACING EXAMPLE:

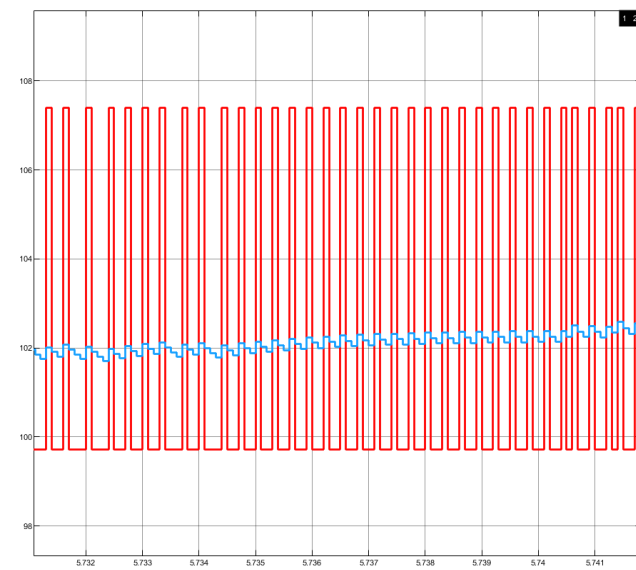
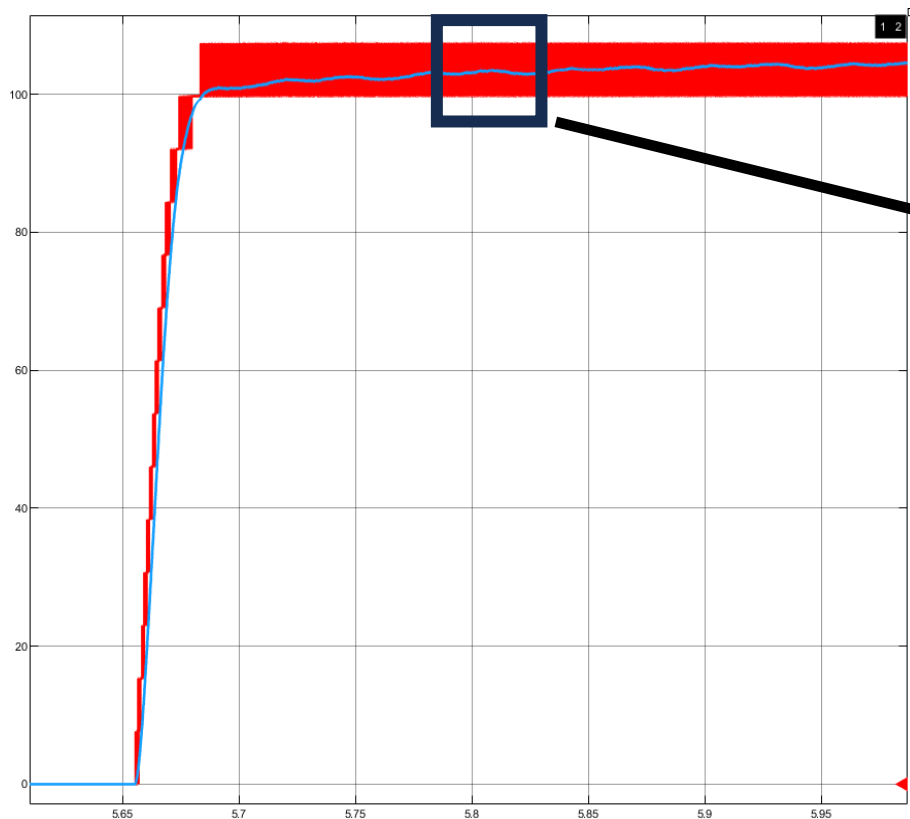
OmegaREF
OmegaMIS



%Desired Taow
Taow=10e-3;

SOME TRACING EXAMPLE:

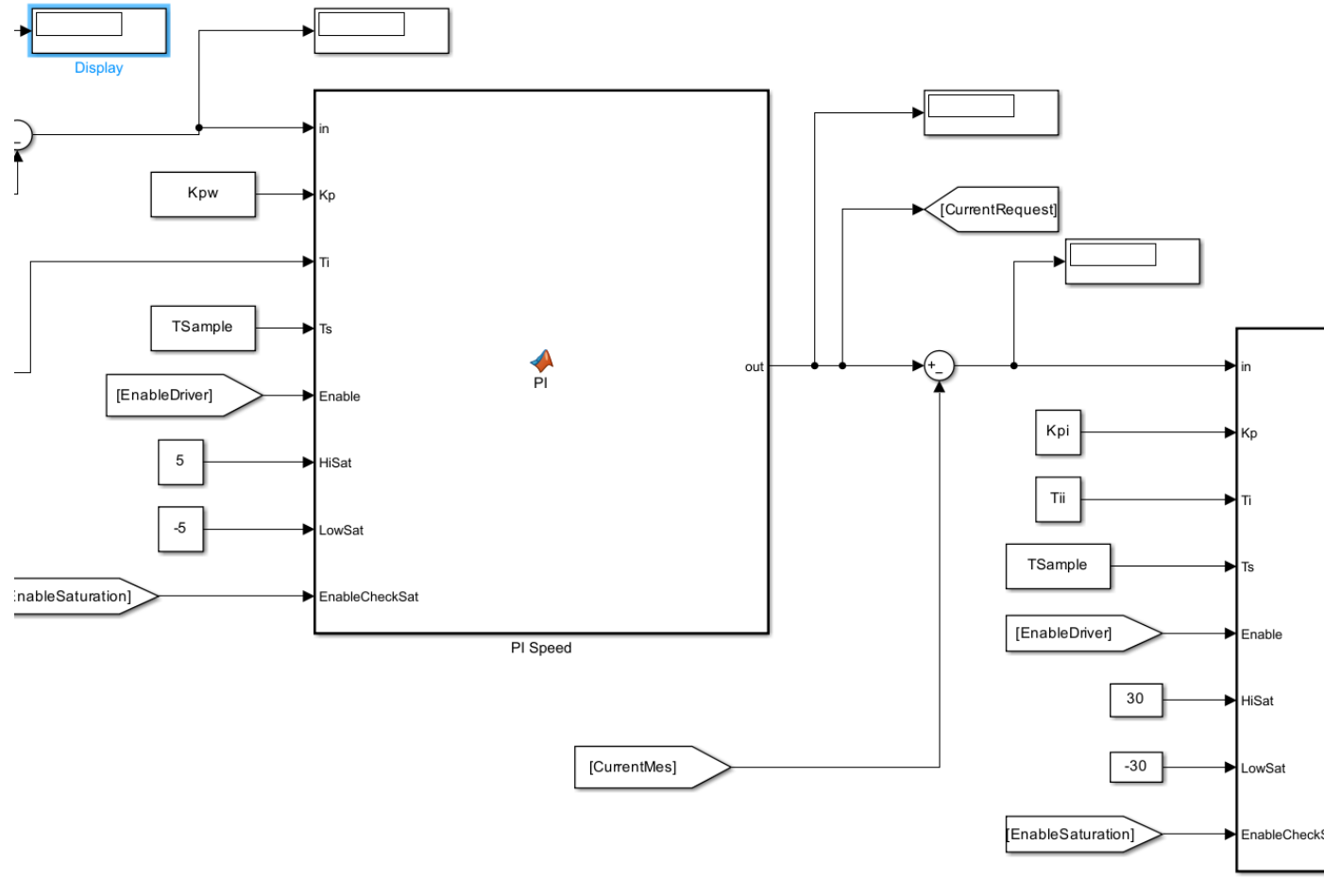
OmegaMis
OmegaFilt



% Omega filter
TaoFilterOmega=0.002;

Simulink connection:

- We can connect a lot of display
- We can change value
- We can use switch
- Ecc...



PROBLEMS:

We cannot see the dynamic behavior of our control; we can observe the steady state values but not the transients.

Example:

with the displays we see that a variable is constantly changing but we cannot understand its trend.

SOLUTION:

We change the way we connect to our microcontroller.

NOW:

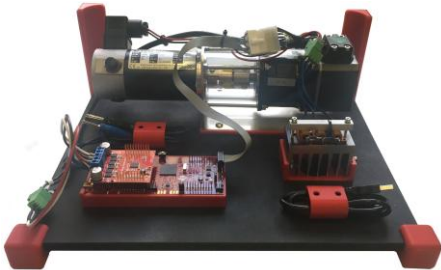


- 1: Code generation
- 2: download
- 3: code execution
- 4: information exchange with Simulink
- 5: Variables update every hundred ms

NEXT

FIRST STEP:

- 1: Code generation
- 2: download
- 3: code execution



SECOND STEP:

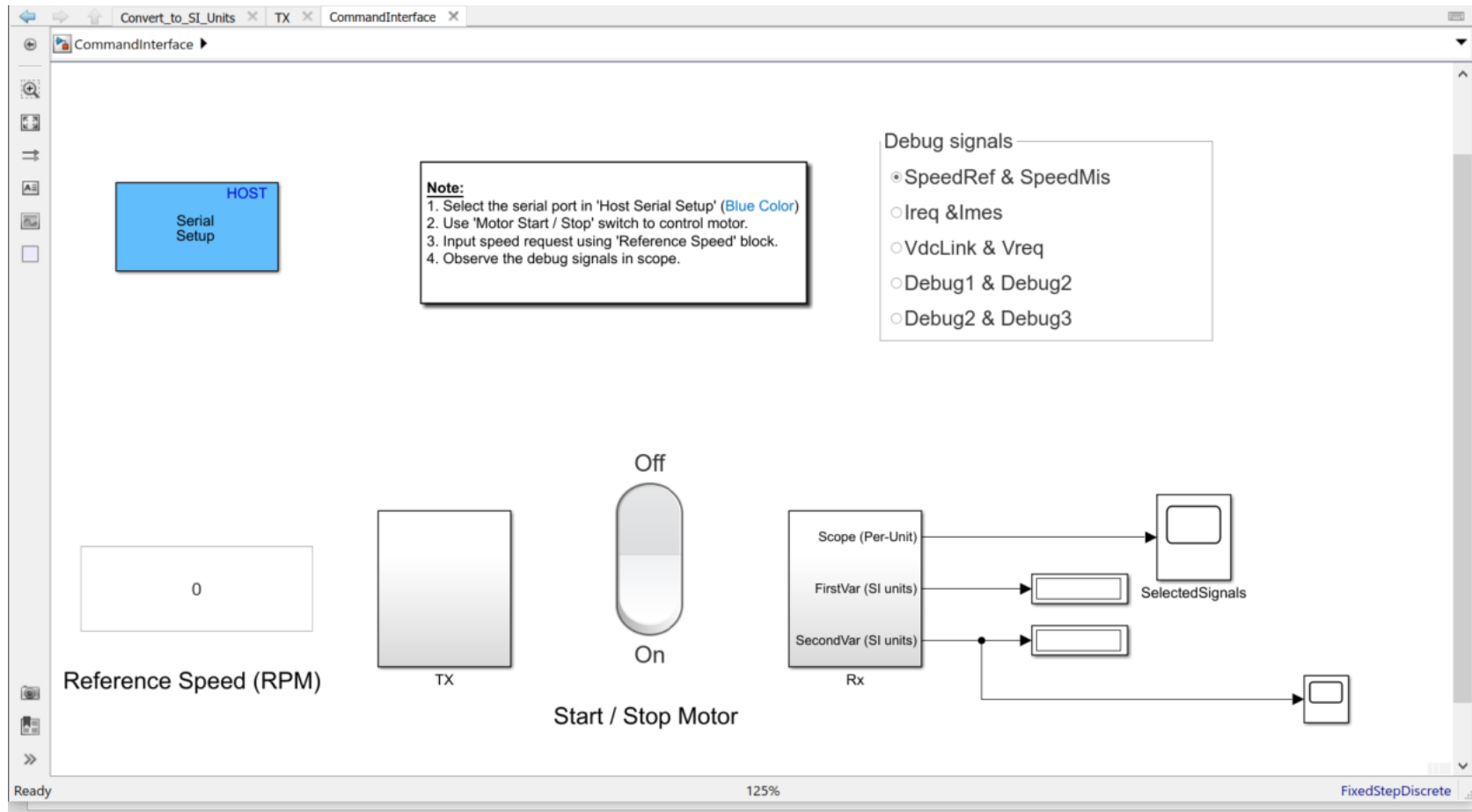
We open a new Simulink Project to exchange data, there is no control but only a communication interface.

Via serial port we ask the microcontroller which variables we want to see (The set of variables should be defined before programming) and we can enable/disable the control and motor

We send the received values to a scope and/or to our matlab workspace



SECOND STEP:



LIMITATION FROM THE PREVIOUS MODE:

- The variables that we can see runtime are limited
- We cannot put as many switches as we want
- **At the same time we can only see two variables**
-

It is useful to do as we did, use the first mode to check the control in a rough way, when we see that everything works, we switch to the new mode.

You can download and open BaseForTracing.slx

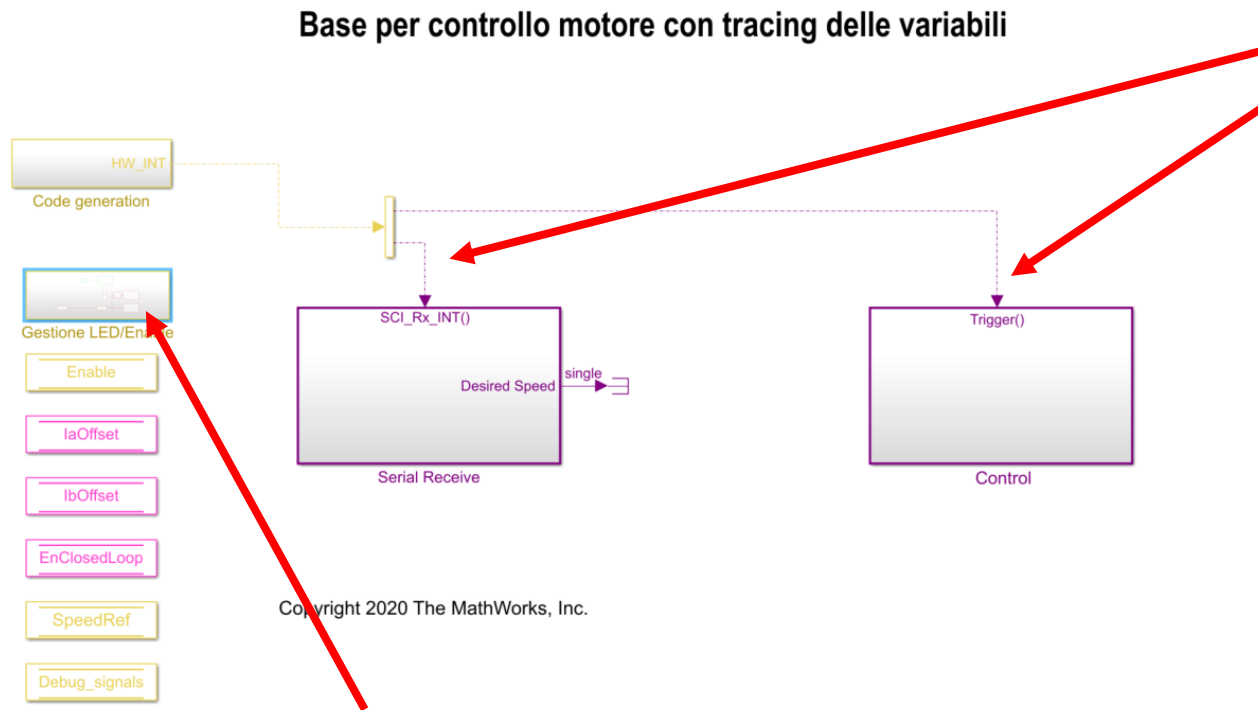
 **Generale** Post File   

 **Nuovo**   Carica   Condividi  Copia collegamento   Tutti i documenti   

Documenti > General > Materiale del corso  > Simulink > BaseModelForTracing 

 Nome 	Data/ora modifica 	Modificato da 	 Aggiungi colonna
  BasePerTracing.slx	Pochi secondi fa	Davide Samorì	
  CommandInterface.slx	Pochi secondi fa	Davide Samorì	

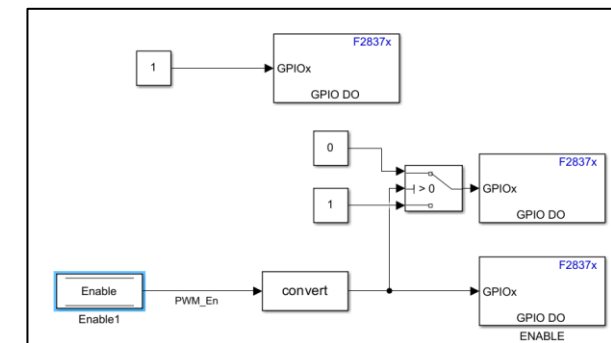
LET'S SEE HOW TO INTEGRATE OUR PROJECT:



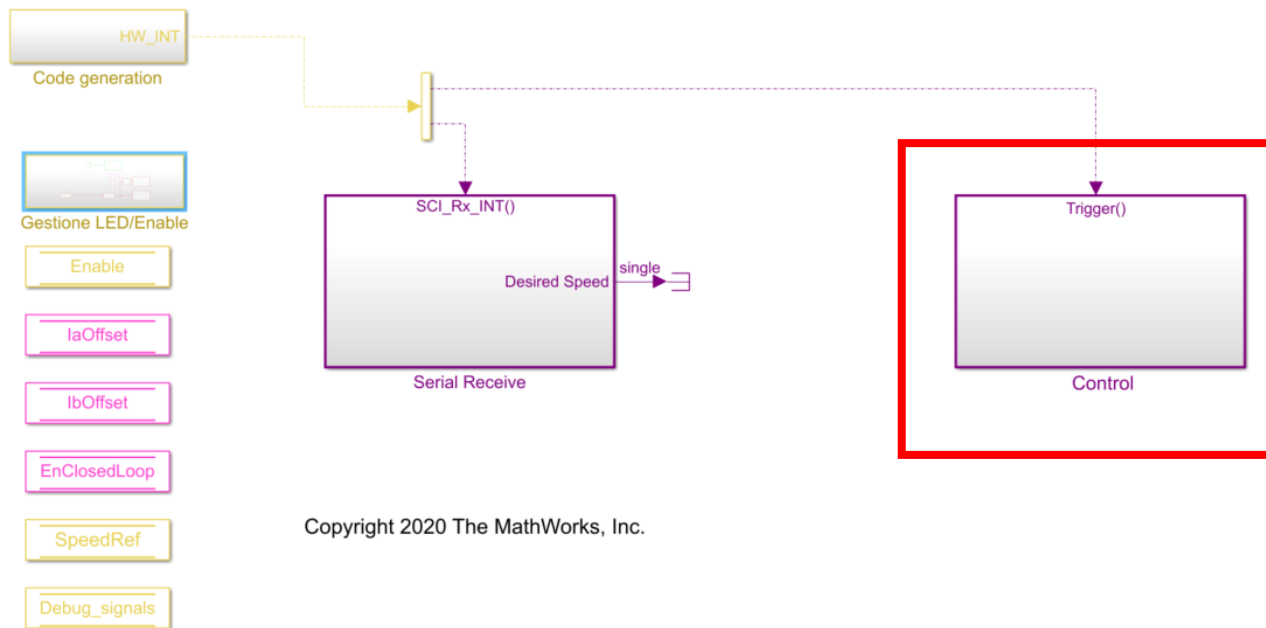
LED and Enable H-Bridge are already here

Now we have two interrupts

- Receive/send data
 - Variable read
 - Enable Control
 - Reference Value
 -
- Our Control Interrupt (Tsample)

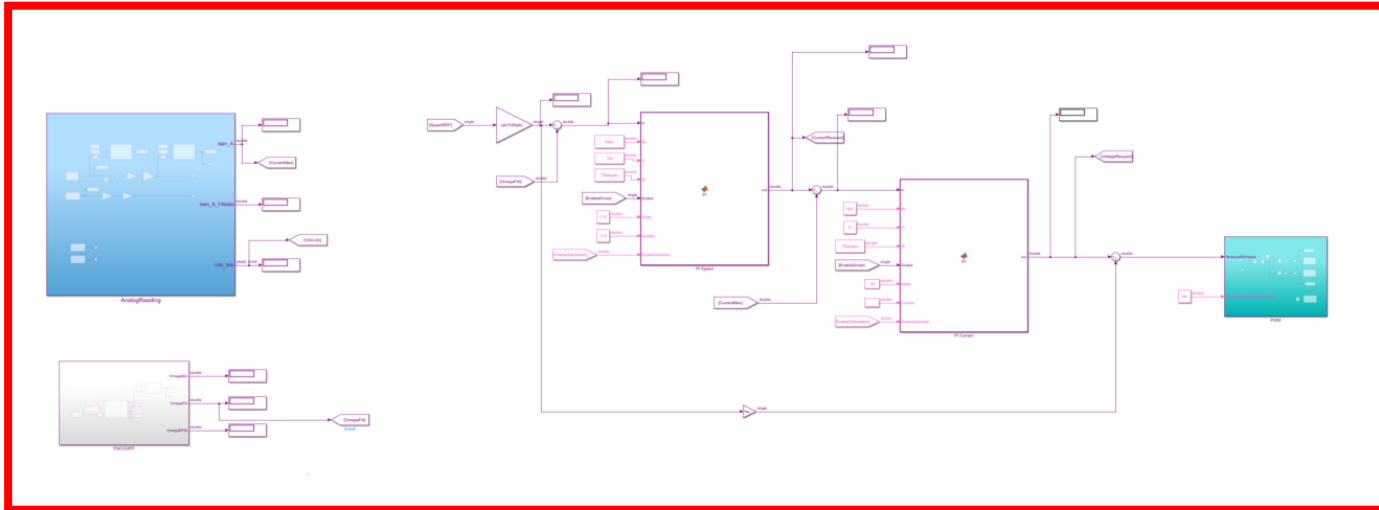
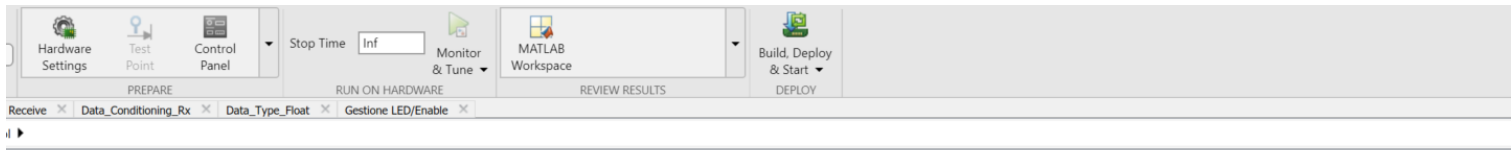


Base per controllo motore con tracing delle variabili

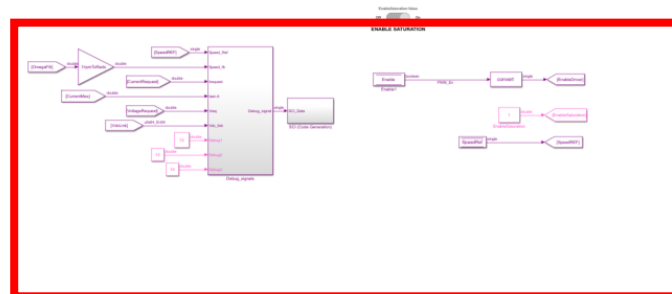


We work only inside this block

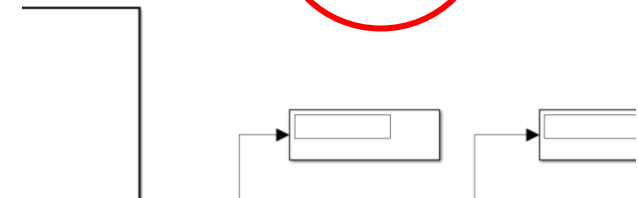
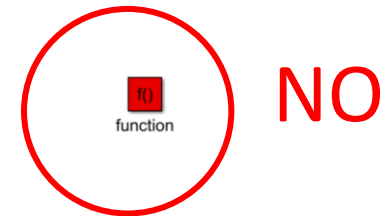
Copyright 2020 The MathWorks, Inc.

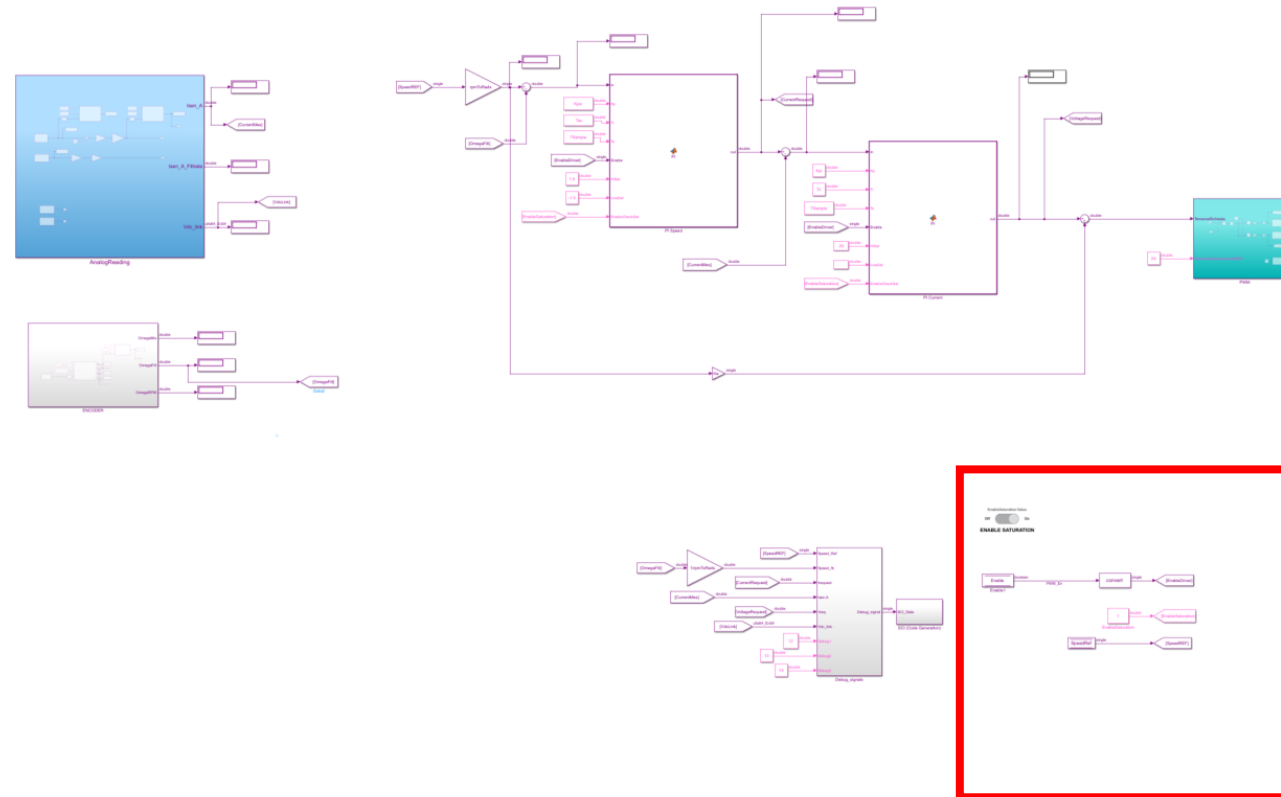
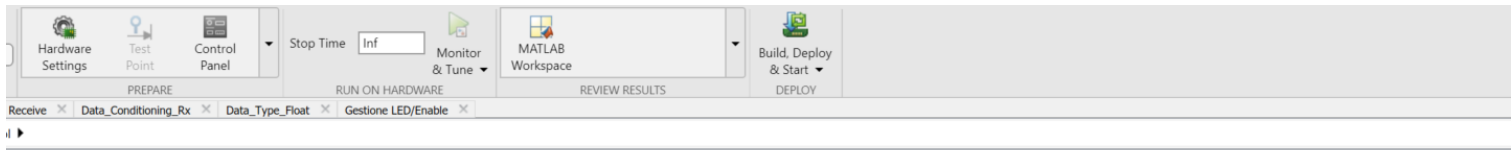


Communication
blocks

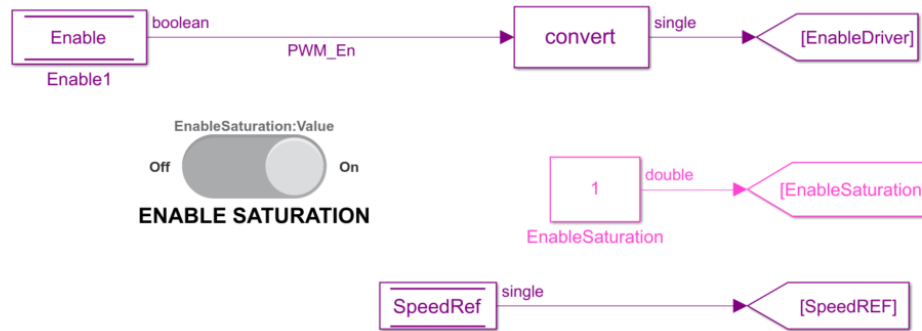


You can copy all your control
here except for the trigger
block





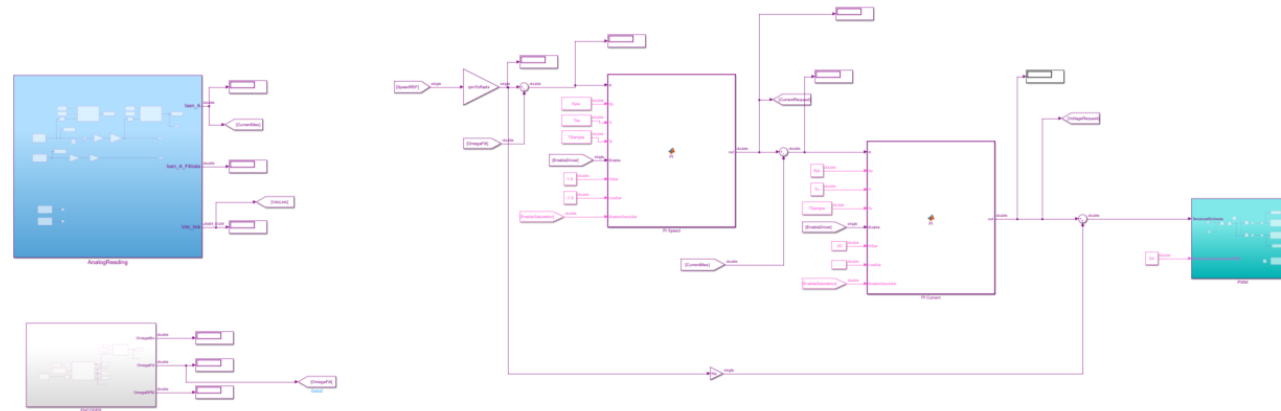
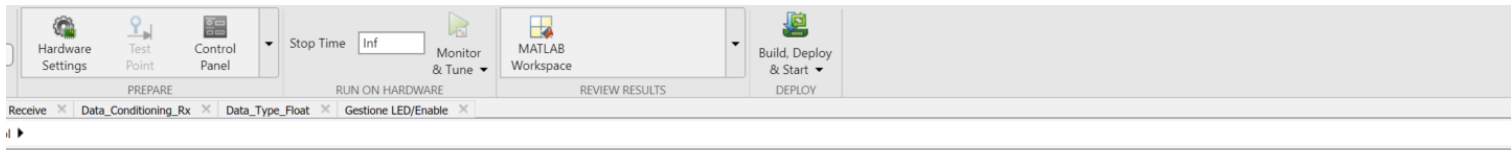
Enable Control and Speed Reference are already available



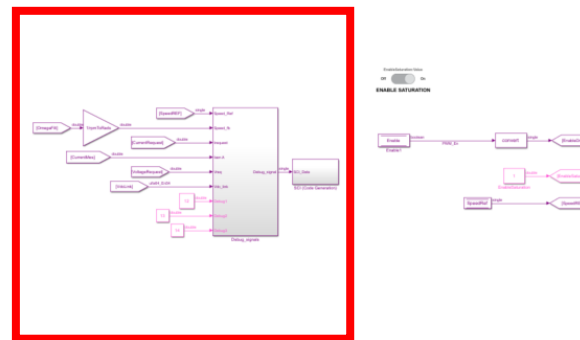
Control enable and speed SetPoint are not decided from this model but receive them from serial communication

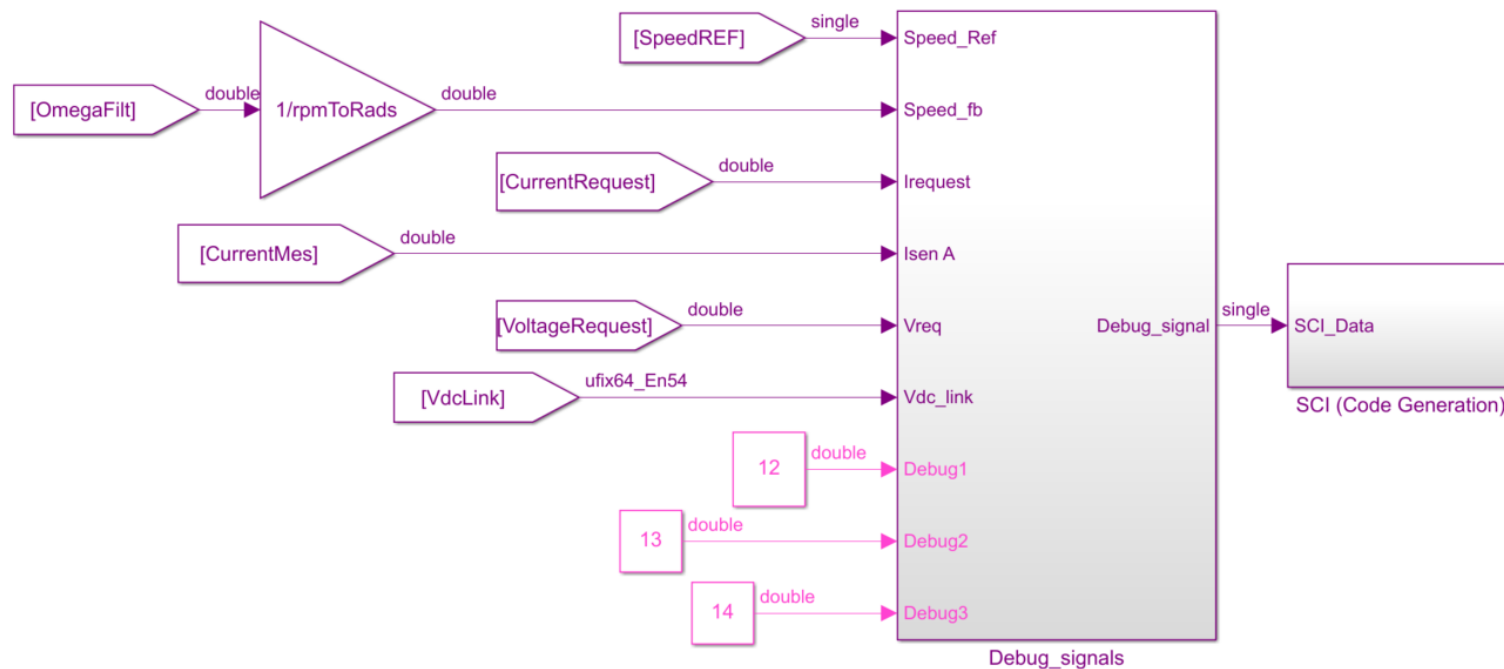
In your control use these Labels:

- EnableDriver
- EnableSaturation
- SpeedREF



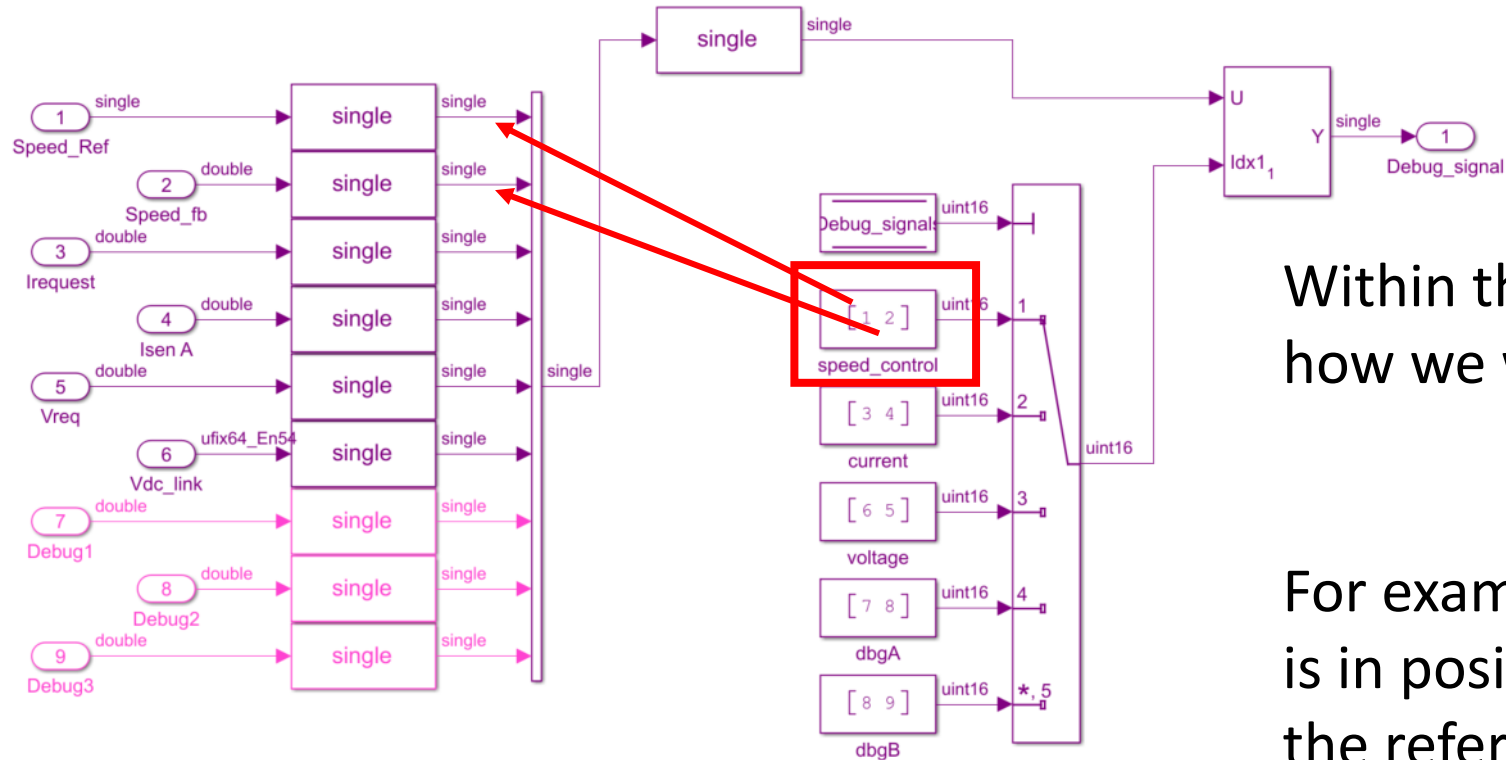
Tracing





Here we connect the variables we want to see :

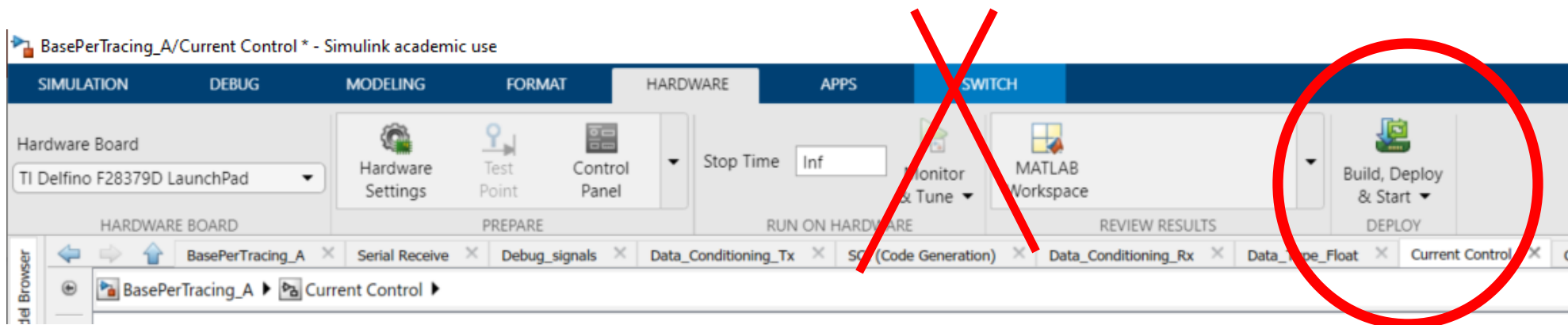
- I have already linked Some of them because they are useful for control visualization
- Some are free
- You can link the variables you want



Within the block we decide how we want them displayed:

For example, when the switch is in position 1 we can display the reference speed and the measured speed.

To Run Your model, click **Build, Deploy & Start**



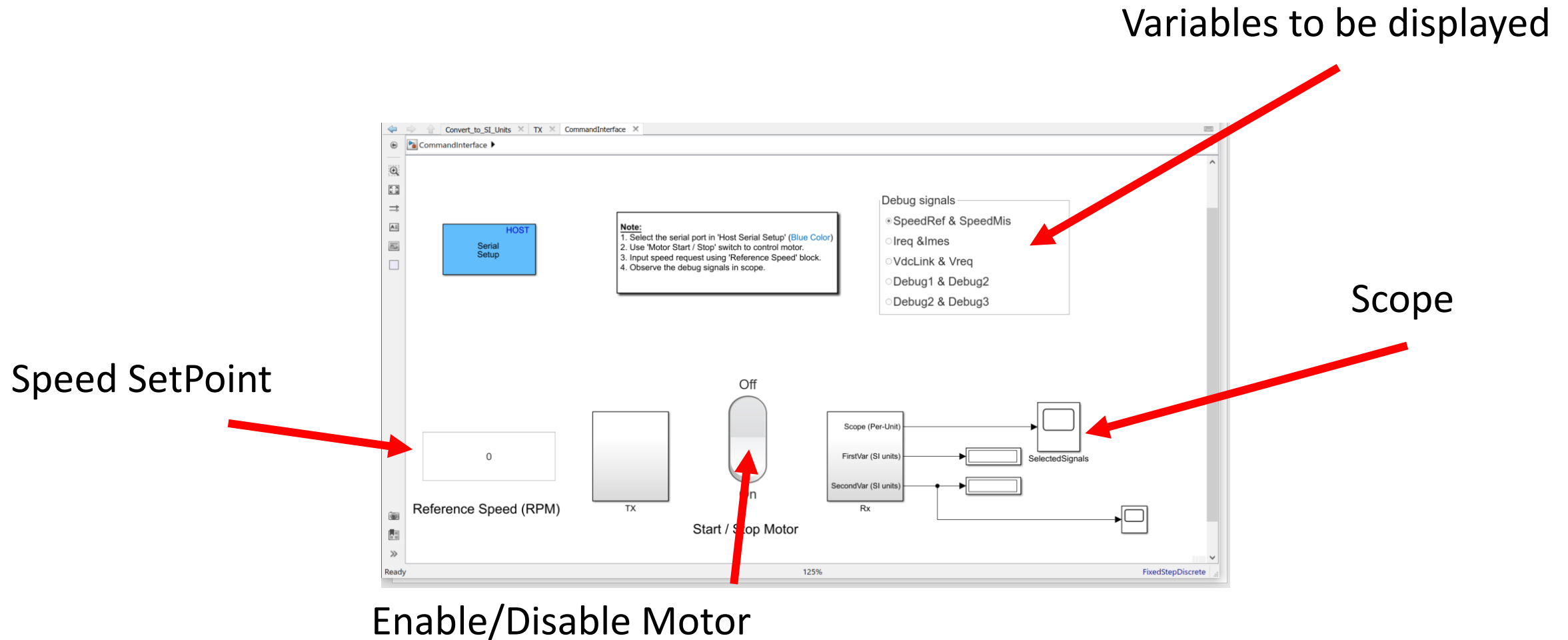
You can download and open **CommandInterface.slx**

Add in your matlab script:

```
comport="COMXX";
```

where XX is your com number

LET'S SEE THE COMMAND INTERFACE



1: Reference Speed=0

2: Press Run

3: Select Debug signal

4: Press ON Button

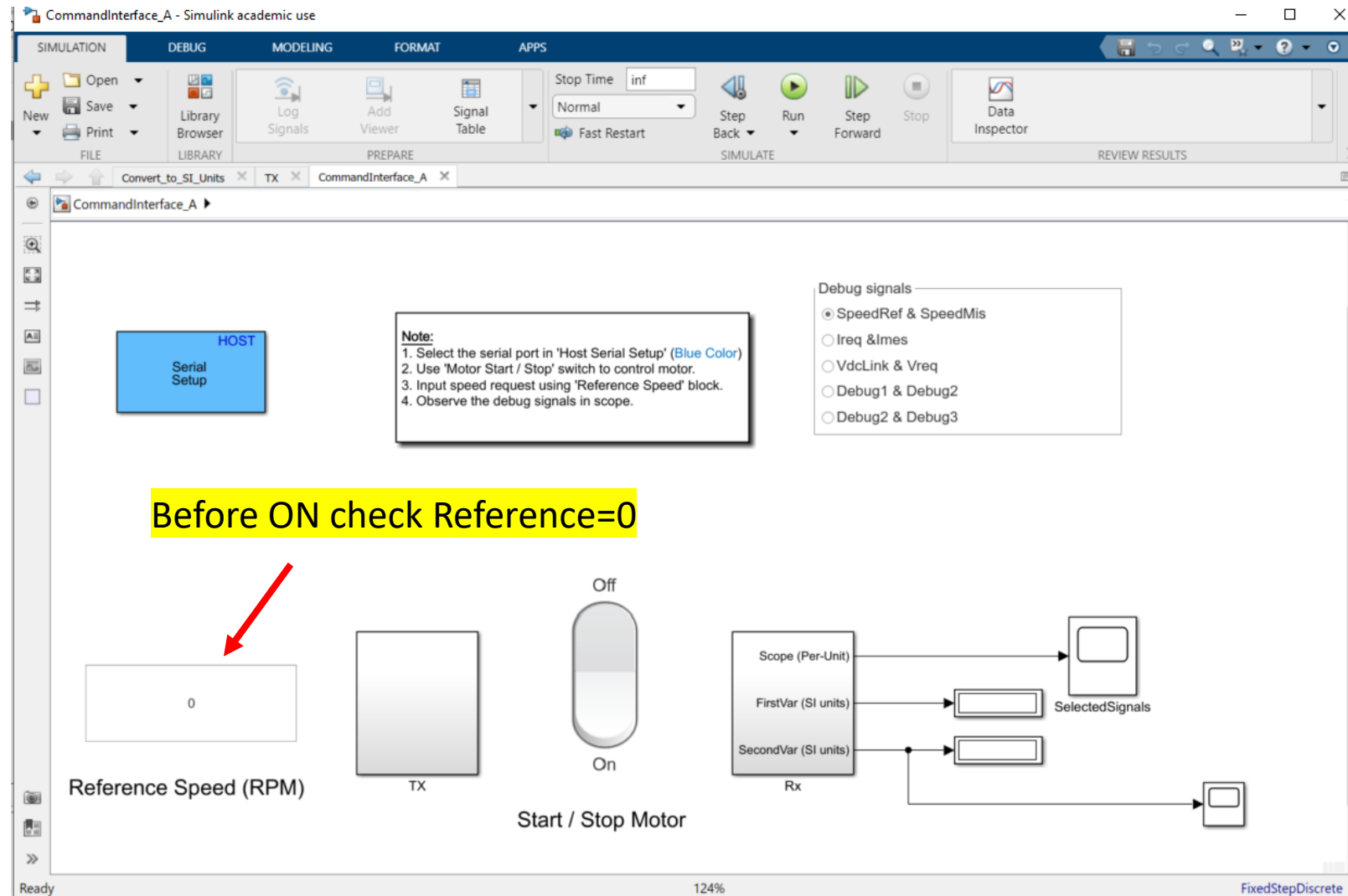
5: Write a
reference speed

6: Reference Speed=0

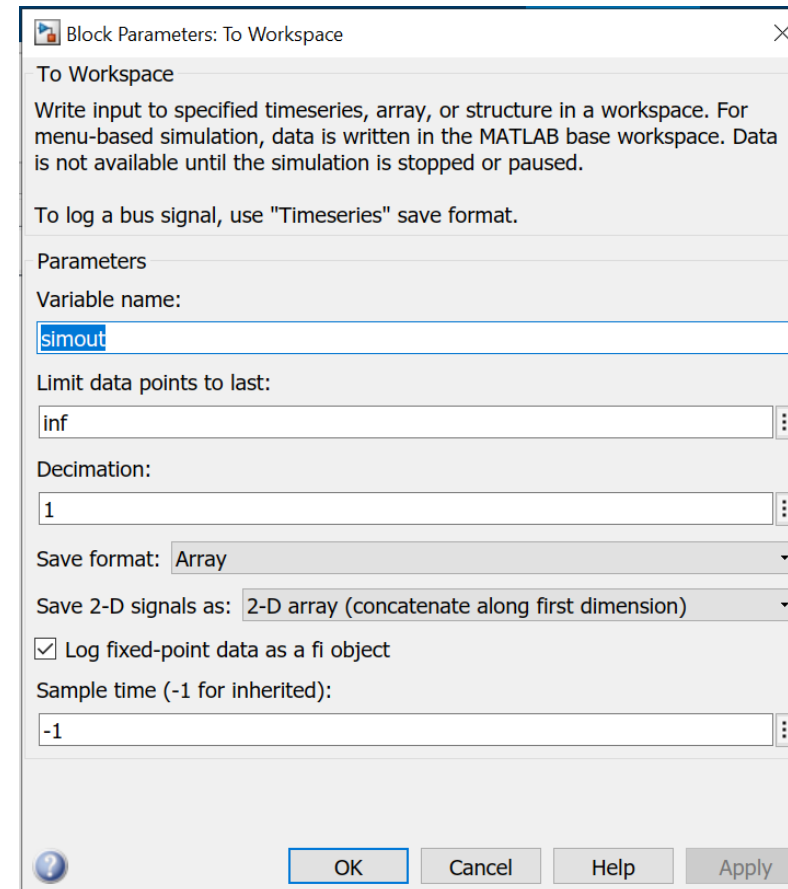
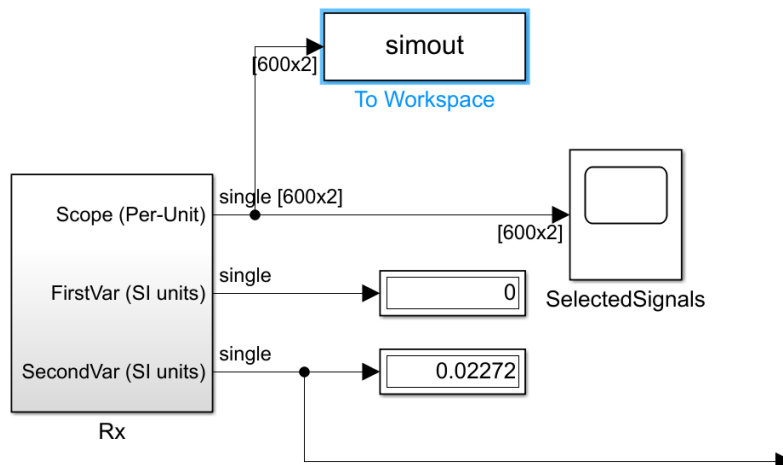
7: Press OFF Button

8: STOP Simulation

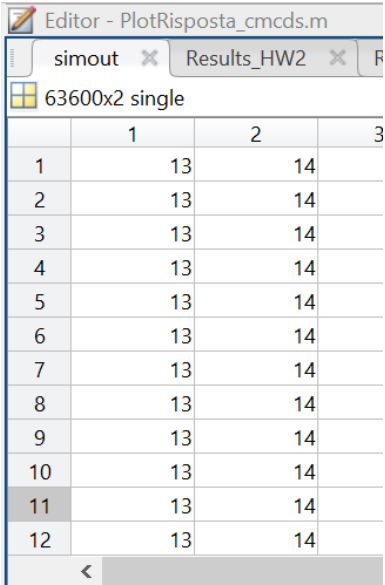
9: Open Scope



To get the acquisitions on matlab you can use a **ToWorkspace** block.



The two selected variables are entered into an array with two columns.



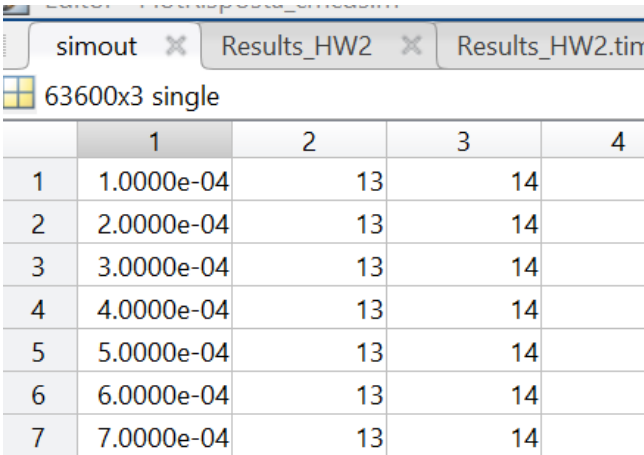
Editor - PlotRisposta_cmcds.m

simout x Results_HW2 x

63600x2 single

	1	2	3
1	13	14	
2	13	14	
3	13	14	
4	13	14	
5	13	14	
6	13	14	
7	13	14	
8	13	14	
9	13	14	
10	13	14	
11	13	14	
12	13	14	

Each row is sent to each TSample, so the time axis for example can be constructed by adding a column with times:



Editor - PlotRisposta_cmcds.m

simout x Results_HW2 x Results_HW2.time

63600x3 single

	1	2	3	4
1	1.0000e-04	13	14	
2	2.0000e-04	13	14	
3	3.0000e-04	13	14	
4	4.0000e-04	13	14	
5	5.0000e-04	13	14	
6	6.0000e-04	13	14	
7	7.0000e-04	13	14	

TIPS:

- When the command window is running, a **lot of data comes inside our matrix**, so you better pause/stop it if you don't have to look at the graphs.
- Simout grows quickly, so you'd better whenever you want to acquire something, **start with the empty matrix** and rename it at the end of the trial.
- **Give sensible names** so that you can process the data at home at your leisure, perhaps match them with the graph screenshots.